FINDS: An N-Body Simulator for Computing the Evolution of Large Schools

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Abstract

Animals travelling in cohesive groups yields many collective and individual benefits, such as security against predators and energy efficiency. The replication of such benefits for analogous, biology-inspired technologies like autonomous drone systems requires keen understanding of the cohesion and evolution of systems of these groups, but these investigations are fundamentally limited to small sizes due to the high computational complexity in computing pairwise dynamics at order N^2. As real schools of fish have been recorded to contain millions of members, FINDS, or Fish INteraction and Dynamics Simulator, was created to efficiently simulate the dynamics of very large schools of fish using common computational techniques employed in N-body simulation. FINDS is implemented as a containerized, modular software suite written predominantly in C (with Python for post-processing) that handles three core simulation processes in order. First, FINDS contains routines for computing the time-derivative of a school, which is a function of the sum of the pairwise interactions for each member of the school. As such, FINDS contains implementations of Brute Force computation, the Barnes-Hut algorithm, and the Fast Multipole Method. Second, FINDS contains custom routines for numerically integrating these systems using adaptive and non-adaptive Runge-Kutta schemes for adaptive time-stepping. Third, FINDS contains a small library of post-processing modules written in Python for stream-lining the analysis of generated data.

Introduction

Synchronous mass movement is a common patten throughout nature, observed most commonly in the movement of smaller animals or prey animals such as birds, fish, bugs, oxen, deer, etc. due to the multitude of benefits that members of such groups gain, such as increased protection from predators, increased energy efficiency due to the motion of others (such as vortical wake), etc. *finish later!!!!!!*

Literature Review and Similar Work

Mathematical Model of Swimmers

FINDS is developed around the far-field source/sink dipole model designed by Mabrouk and Floryan for modelling how passive hydrodynamic interactions at far distances effects the evolution of a greater school [1, 2]. Under this model, each fish i is defined as a dipole consisting of a fluid source at the head of the fish $\hat{\mathbf{x}}_{f,i}$, and a fluid sink at the tail $\hat{\mathbf{x}}_{b,i}$, separated by the length of the fish ℓ_i. The source and the sink of each fish are defined to have the same volumetric flow rate, σ_i, to ensure the conservation of fluid by making the sinks consume exactly the same amount of fluid as the sources. For simplicity, the source and sink (or "front" and "back") of each fish are called its "features."

Each fish is defined as an eight-dimensional vector consisting of four variables: $\vec{\mathbf{x}}_{c,i}$, the three-dimensional center-of-mass position vector, $\hat{\mathbf{n}}_i$, the three-dimensional orientation unit vector, ℓ_i, and σ_i. Together, the collection of each of these fish vectors at a given time forms the state vector $\vec{\mathbf{X}}$.

In particular, $\vec{\mathbf{x}}_{c,i}$ and $\hat{\mathbf{n}}_i$ are dynamic and ℓ_i and σ_i are constant for each fish i. This formulism leaves the translational and rotational dynamics for each fish as the first-order differential equations concerning $\vec{\mathbf{x}}_{c,i}$ and $\hat{\mathbf{n}}_i$ respectively. With this, the derivatives of each are defined as

$$\frac{\mathrm{d}\vec{\mathbf{x}}_{c,i}}{\mathrm{d}t} = \frac{1}{2}\left(\frac{\mathrm{d}\vec{\mathbf{x}}_{f,i}}{\mathrm{d}t} + \frac{\mathrm{d}\vec{\mathbf{x}}_{b,i}}{\mathrm{d}t}\right) \tag{1}$$

$$\frac{\mathrm{d}\hat{\mathbf{n}}_i}{\mathrm{d}t} = \frac{\Delta\vec{\mathbf{v}}_i + 2\lambda_i\hat{\mathbf{n}}_i}{\ell_i} \tag{2}$$

where

$$\lambda_i = \frac{\Delta\vec{\mathbf{v}}_i \cdot \hat{\mathbf{n}}}{2} \tag{3}$$

is a Lagrange multiplier to ensure that ℓ_i is kept constant, and

$$\Delta\vec{\mathbf{v}}_i = \frac{\mathrm{d}\vec{\mathbf{x}}_{f,i}}{\mathrm{d}t} - \frac{\mathrm{d}\vec{\mathbf{x}}_{b,i}}{\mathrm{d}t}. \tag{4}$$

With this, the translational and rotational dynamics for each fish are variably dependent on the translational derivatives of

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the source and sink for each fish, and this reduces the problem of modelling fish to an analogous problem of simple Laplacian particle dynamics based on treating the sources and sinks as individual particles with charge σ_i. Under this lens, the potential function resulting from a particle is the charge multiplied by Green's function for the three-dimensional Laplace equation,

$$\Phi_i(r) = \sigma_i G(r) = -\frac{\sigma_i}{4\pi r}. \tag{5}$$

However, as this model is designed for far-field interactions, it becomes unphysical for small distances, as infinitely small distances produce infinitely large potential. To solve this problem, Mabrouk and Floryan reformulated this potential with a regularization parameter ε defined as

$$\Phi_{i,\varepsilon}(r) = \Phi_i(r)\,\mathrm{erf}(r/\varepsilon). \tag{6}$$

From the potential functions, the velocity field resulting from any source or sink is defined as the gradient of the potential function, and for simplicity, the displacement-dependent vector $\vec{\mathbf{k}}$ can be defined to represent the gradient of the Laplacian interaction kernel $1/r$,

$$\vec{\mathbf{k}}(\vec{\mathbf{r}}) = \boldsymbol{\nabla}\left(-\frac{1}{r}\right) = \frac{\vec{\mathbf{r}}}{r^3} \tag{7}$$

$$\vec{\mathbf{k}}_\varepsilon(\vec{\mathbf{r}}) = \boldsymbol{\nabla}\left(-\frac{\mathrm{erf}(r/\varepsilon)}{r}\right) = \vec{\mathbf{k}}(\vec{\mathbf{r}})\xi_\varepsilon(r) \tag{8}$$

where $\xi_{i,\varepsilon}(r)$ is the regularization coefficient

$$\xi_\varepsilon(r) = \mathrm{erf}(r/\epsilon) - \frac{2r}{\varepsilon\sqrt{\pi}}\exp\left(-\frac{r^2}{\varepsilon^2}\right). \tag{9}$$

The velocity fields are therefore

$$\vec{\mathbf{u}}_i(\vec{\mathbf{r}}) = \frac{\sigma_i}{4\pi}\vec{\mathbf{k}}(\vec{\mathbf{r}}) \tag{10}$$

$$\vec{\mathbf{u}}_{i,\varepsilon}(\vec{\mathbf{r}}) = \frac{\sigma_i}{4\pi}\vec{\mathbf{k}}_\varepsilon(\vec{\mathbf{r}}) \tag{11}$$

for the unregularized and regularized velocity fields respectively.

With the velocity fields well-defined, the velocity of a given source or sink α can be defined generally as

$$\frac{\mathrm{d}\vec{\mathbf{x}}_{\alpha,i}}{\mathrm{d}t} = U_i\hat{\mathbf{n}}_i + \sum_{i\neq j}^{N}\frac{\sigma_j}{4\pi}\vec{\mathbf{K}}_{\alpha,ij}. \tag{12}$$

where the first term is the *internal* velocity contribution and the second term is the *external* velocity contribution. In detail,

$$U_i = \frac{\sigma_i}{4\pi\ell_i^2} \tag{13}$$

is the self-propelled speed resulting from the interaction between any source-sink pair of a dipole, and the second term represents the sum of the velocity contribution to a source or sink from a dipole, where $\vec{\mathbf{K}}_{\alpha,ij}$ represents the composite interaction vector between feature α of fish i with both the source f and the sink b of fish j,

$$\vec{\mathbf{K}}_{\alpha,ij} = \vec{\mathbf{k}}(\vec{\mathbf{r}}_{\alpha f}) - \vec{\mathbf{k}}(\vec{\mathbf{r}}_{\alpha b}). \tag{14}$$

where $\vec{\mathbf{r}}_{\alpha\beta}$ is the displacement between feature α of fish i and feature β of fish j. In the case of regularization, the interaction function $\vec{\mathbf{k}}$ is simply replaced with $\vec{\mathbf{k}}_\varepsilon$.

Derivative Computation

As stated previously, the modelling of fish is an N-body problem where the algorithmic complexity of computing a derivative is, through Brute-Force calculations, of order $\mathcal{O}(N^2)$. To reduce this computation time, two tree-based approximation algorithms are implemented: the Barnes-Hut algorithm and the Fast Multipole Method, of orders $\mathcal{O}(N\log N)$ and $\mathcal{O}(N)$ respectively.

Brute force computation and the Barnes-Hut approximation are implemented directly into the FINDS code-base, and as such, the user has the ability to switch between unregularized and regularized calculations with ease. However, the Fast-Multipole Method is computed externally using Greengard et al.'s Fast Multipole Method library FMM3D, which will only compute the standard Laplacian gradient $\vec{\mathbf{u}}_i$, so the Fast-Multipole Method cannot be used with regularization.

Barnes-Hut Approximation

Fast Multipole Method

Numerical Integration

FINDS implements multiple

System Creation

Database Management

Validation

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Performance Evaluation

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Examples

Limitations

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Future Work

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Acknowledgements

In the development of FINDS, Alvin Ng's `grav_sim` was used extensively as a reference for implementing a fast N-body simulator in C, particularly in implementing of the Barnes-Hut algorithm. In addition, the direct advice of Alvin Ng was instrumental to the implementation of linear octrees and parallelization.

Furthermore, generative artificial intelligence (Anthropic's Sonnet 5 Claude, OpenAI's ChatGPT-5.5 and Google's Gemini 3.3) were used heavily for initial code generation before being thoroughly tested and validated to ensure correctness. Artificial intelligence was not used in developing the content of this report whatsoever, including the abstract, section text, figures, and captions.

References

[1] Mohamed Niged Mabrouk and Daniel Floryan. "Group cohesion and passive dynamics of a pair of inertial swimmers with three-dimensional hydrodynamic interactions". In: *Bioinspiration and Biomimetics* 20.1 (Nov. 2024), p. 016014. DOI: 10.1088/1748-3190/ad936d. URL: https://doi.org/10.1088/1748-3190/ad936d.

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Appendix